

ISyE 6525 HW2 - Question 1

(a) Between-class variance

Let $N = mn$ be the total number of pixels. Define the two class-indicator functions by

$$I_1(x_{ij}) = \begin{cases} 1, & x_{ij} \leq t, \\ 0, & x_{ij} > t, \end{cases} \quad I_2(x_{ij}) = \begin{cases} 1, & x_{ij} > t, \\ 0, & x_{ij} \leq t. \end{cases}$$

Therefore,

$$I_1(x_{ij}) + I_2(x_{ij}) = 1.$$

The class weights and class means can be written as

$$\omega_k = \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n I_k(x_{ij}), \quad \mu_k = \frac{\sum_{i=1}^m \sum_{j=1}^n x_{ij} I_k(x_{ij})}{\sum_{i=1}^m \sum_{j=1}^n I_k(x_{ij})}, \quad k \in \{1, 2\}.$$

Hence,

$$\sum_{i=1}^m \sum_{j=1}^n I_k(x_{ij}) = N\omega_k, \quad \sum_{i=1}^m \sum_{j=1}^n x_{ij} I_k(x_{ij}) = N\omega_k\mu_k.$$

The total mean is therefore

$$\begin{aligned} \mu_T &= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n x_{ij} \\ &= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n x_{ij} (I_1(x_{ij}) + I_2(x_{ij})) \\ &= \omega_1\mu_1 + \omega_2\mu_2. \end{aligned}$$

Expanding the total variance gives

$$\begin{aligned}
\sigma_T^2 &= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n (x_{ij} - \mu_T)^2 \\
&= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n x_{ij}^2 - \mu_T^2.
\end{aligned} \tag{1}$$

For each class,

$$\sigma_k^2 = \frac{\sum_{i=1}^m \sum_{j=1}^n I_k(x_{ij})(x_{ij} - \mu_k)^2}{\sum_{i=1}^m \sum_{j=1}^n I_k(x_{ij})}.$$

Thus, the within-class variance can be expressed using the indicators as

$$\begin{aligned}
\sigma_w^2 &= \omega_1 \sigma_1^2 + \omega_2 \sigma_2^2 \\
&= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n [I_1(x_{ij})(x_{ij} - \mu_1)^2 + I_2(x_{ij})(x_{ij} - \mu_2)^2] \\
&= \frac{1}{N} \left[\sum_{i=1}^m \sum_{j=1}^n x_{ij}^2 (I_1(x_{ij}) + I_2(x_{ij})) - 2\mu_1 \sum_{i=1}^m \sum_{j=1}^n x_{ij} I_1(x_{ij}) - 2\mu_2 \sum_{i=1}^m \sum_{j=1}^n x_{ij} I_2(x_{ij}) \right. \\
&\quad \left. + \mu_1^2 \sum_{i=1}^m \sum_{j=1}^n I_1(x_{ij}) + \mu_2^2 \sum_{i=1}^m \sum_{j=1}^n I_2(x_{ij}) \right] \tag{2} \\
&= \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^n x_{ij}^2 - \omega_1 \mu_1^2 - \omega_2 \mu_2^2.
\end{aligned}$$

(1) - (2), it follows that

$$\begin{aligned}
\sigma_b^2 &= \sigma_T^2 - \sigma_w^2 \\
&= \omega_1 \mu_1^2 + \omega_2 \mu_2^2 - \mu_T^2 \\
&= \omega_1 \mu_1^2 + \omega_2 \mu_2^2 - (\omega_1 \mu_1 + \omega_2 \mu_2)^2 \\
&= \omega_1(1 - \omega_1) \mu_1^2 + \omega_2(1 - \omega_2) \mu_2^2 - 2\omega_1 \omega_2 \mu_1 \mu_2.
\end{aligned}$$

Since $\omega_1 + \omega_2 = 1$, we have

$1 - \omega_1 = \omega_2$ and $1 - \omega_2 = \omega_1$. Hence,

$$\begin{aligned}
\sigma_b^2 &= \omega_1 \omega_2 \mu_1^2 + \omega_1 \omega_2 \mu_2^2 - 2\omega_1 \omega_2 \mu_1 \mu_2 \\
&= \omega_1 \omega_2 (\mu_1 - \mu_2)^2
\end{aligned}$$

Therefore,

$$\sigma_b^2(t) = \omega_1(t)\omega_2(t) (\mu_1(t) - \mu_2(t))^2.$$

(b) Endpoint behavior

At either endpoint, the threshold produces a degenerate partition in which one class contains all the pixels and the other class is empty. Thus, at $t = 0$ or $t = 255$, one of the class weights is zero and $\omega_1(t)\omega_2(t) = 0$.

Consequently,

$$\sigma_b^2(t) = \omega_1(t)\omega_2(t) (\mu_1(t) - \mu_2(t))^2 = 0.$$

For an interior threshold, the two factors in $\sigma_b^2(t)$ represent competing effects. The factor $\omega_1(t)\omega_2(t)$ favors a balanced partition. To see this using the Cauchy-Schwarz inequality,

$$(\omega_1 + \omega_2)^2 \leq (1^2 + 1^2) (\omega_1^2 + \omega_2^2).$$

Since $\omega_1 + \omega_2 = 1$, this gives

$$\omega_1^2 + \omega_2^2 \geq \frac{1}{2}.$$

Therefore,

$$\begin{aligned} \omega_1\omega_2 &= \frac{(\omega_1 + \omega_2)^2 - (\omega_1^2 + \omega_2^2)}{2} \\ &\leq \frac{1 - \frac{1}{2}}{2} = \frac{1}{4}. \end{aligned}$$

Equality in Cauchy-Schwarz holds when (ω_1, ω_2) is proportional to $(1, 1)$. Together with $\omega_1 + \omega_2 = 1$, this implies

$$\omega_1 = \omega_2 = \frac{1}{2}.$$

On the other hand, the factor $(\mu_1(t) - \mu_2(t))^2$ favors a threshold that produces classes with strongly separated mean intensities. A threshold near an extreme may isolate a small group of very dark or very bright pixels, giving well-separated class means but a small value of $\omega_1\omega_2$. A threshold closer to the center may balance the class sizes, but it can cut through a concentrated histogram mode and reduce the separation between the class means. Therefore, Otsu's method selects the threshold that gives the best product of class-size balance and class-mean separation.